Environmental Suitability of *Acacia catechu* Across India

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Statistical Ecology

November 1, 2026

Preface

I used AI language tools (a locally run LLM) to edit the wording of this report. They did not produce the data, the model run or the results. Every number comes from my saved model run, and I checked each one against the output files. The interpretations of the charts, maps and graphs are my own.

I have also used Python's elapid model as a substitute for Maxent which uses Java.

Table of Contents

PREFACE **2**

1 INTRODUCTION **4**

2 OCCURRENCE DATA AND STUDY AREA **4**

3 ENVIRONMENTAL DATA AND PREPARATION **6**

3.1 Climate, terrain and vegetation data 6

3.2 Grid alignment and predictor screening 6

4 MODEL CONSTRUCTION AND EVALUATION **6**

4.1 Calibration area and background 6

4.2 Model selection and test region 7

5 RESULTS AND ECOLOGICAL INTERPRETATION **8**

5.1 Variables that contribute most 8

5.2 AUC, strengths, weaknesses and overfitting 11

5.3 What the map shows 12

6 IMPLICATIONS FOR CONSERVATION AND RISK **14**

6.1 Conservation planning 14

6.2 Climate-change studies 15

6.3 Invasive-species risk 15

7 LIMITATIONS AND NEXT STEPS **15**

8 CONCLUSION **16**

REFERENCES **17**

APPENDIX A PREDICTOR TABLE **21**

APPENDIX B CLIPPED PREDICTOR MAPS **22**

APPENDIX C SUPPLEMENTARY GENERATED FIGURES **23**

1 Introduction

Species distribution models link recorded locations to environmental conditions and then they rank other places with similar conditions (Elith & Leathwick, 2009). For *Acacia catechu* in India, the ranking can show where botanical surveys would be most informative. One caveat is that the records carry no planting status, so the model cannot separate planted trees from wild ones.

Acacia catechu as chosen for three reasons. The given data holds 1,796 records across India, enough to reserve a whole region for testing. Secondly, the tree is a deciduous, multipurpose species of seasonally dry South Asian region (ICFRE, n.d.; Kew Science, 2026), so climate and vegetation should matter for it and finally it is also both wild and widely planted, which makes it a useful case for studying sampling bias.

I built a presence–background MaxEnt model from occurrence cells and climate, terrain and vegetation layers. This report describes the data preparation, tests transfer to a test region and interprets the map and response curves.

Plants of the World Online lists *Senegalia catechu* as the accepted name and *Acacia catechu* as a synonym (Kew Science, 2026). But I have retained the name *Acacia catechu* to match the occurrence data and project files.

Earlier work covers biomass and forest use (Deb et al., 2015; Orwa et al., 2009) and seed traits and germination (Jha, 1995; Jhilta et al., 2018). It also covers regeneration and management (Negi et al., 1996; Raizada & Juyal, 2012) and seedling water relations (Puri et al., 1994). Grubb's (1977) regeneration niche reminds me that suitable conditions do not guarantee recruitment. Singh et al. (1974) describe regional vegetation history, and Sourabh et al. (2023) modelled the species' habitat in India. Neither study validates my map.

2 Occurrence data and study area

The course workbook held 1,796 records. The coordinate audit found no invalid numeric coordinates or global-range flags, and all records fell inside the Survey of India boundary. Four were flagged for review because they lay close to the boundary. After cleaning, one record was kept per 30-arc-second cell, leaving 1,279 cells; 516 additional records shared a cell with a kept record, and one record lacked predictor values. These checks found no flagged coordinate-range problems and removed duplicate-cell weighting, but they cannot establish exact positional accuracy, taxonomic identity or wild status.

Records were thinned to one record per cell as the predictor grid is about 1 km wide, so two records in one cell carry identical environmental values. Keeping both would count the same conditions twice. To test this choice, I also thinned to 5 km and 10 km. Test-region AUC stayed near 0.72 (0.721 and 0.720, against 0.729 at 1 km), while omission rose from 3.0% to 5.1% and 7.1%. Larger distances cut the training presences from 941 to about 559 and 401 without improving the score, so I kept 1 km. Thinning at 1 km leaves the wider clustering from uneven access in the data.

The records cluster in northern and central India (Figure 1). This pattern reflects where the species grows and where people have searched. The workbook gives no survey date, effort measure or coordinate uncertainty, so the model may learn search effort as well as habitat.

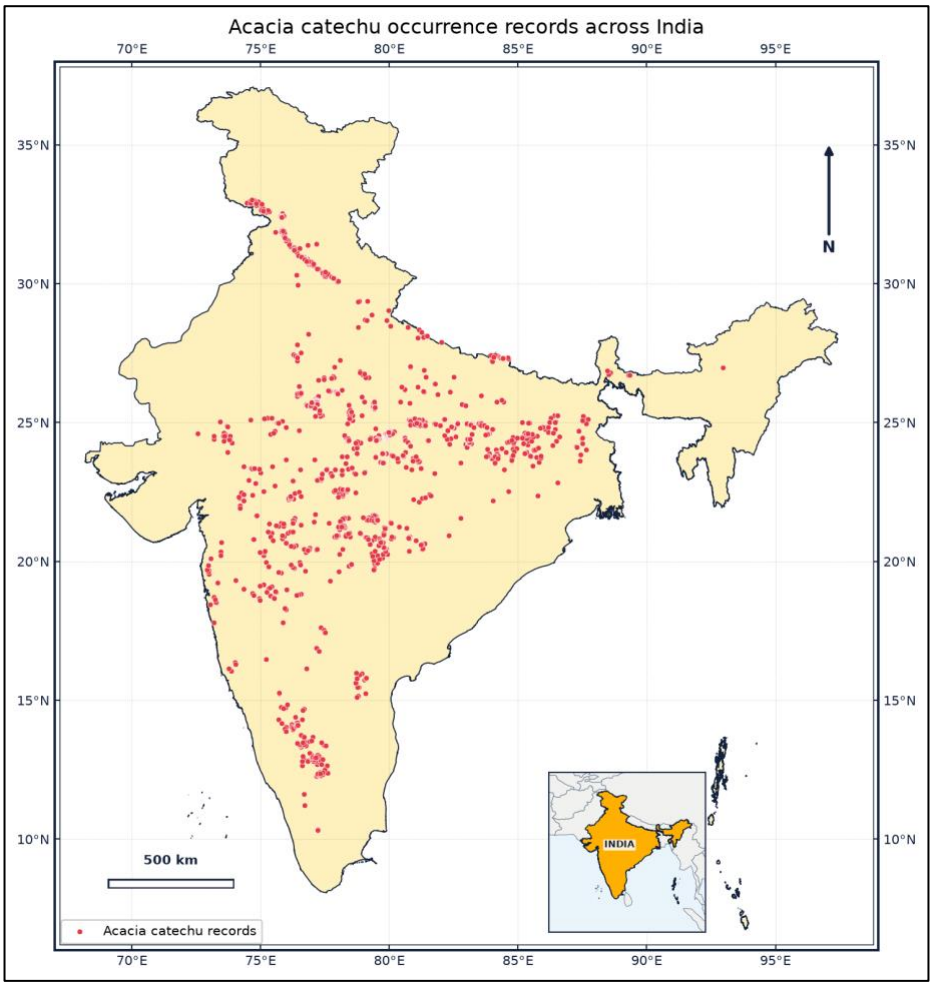


Figure 1. Retained occurrence cells across India. Source: course-provided species workbook and Survey of India boundary; map prepared from the saved project run.

3 Environmental data and preparation

3.1 Climate, terrain and vegetation data

The data used to build the model comprised of the 19 WorldClim 2.1 bioclimatic variables, SRTM-derived elevation and annual mean NDVI from Sentinel-2 imagery processed in Google Earth Engine. WorldClim describes 1970–2000 climate. The NDVI composite covers January to December 2025 and uses Cloud Score+ (cs_cdf ≥ 0.6) to remove cloudy observations (Fick & Hijmans, 2017; Google, n.d.; Google Earth Engine, 2026; WorldClim, 2026). The model therefore pairs recent greenness with climate from decades earlier, so the fitted links mix two time periods and thus is a limitation.

The NDVI composite was sampled on the 30-arc-second target grid rather than explicitly averaged from the native 10-metre pixels within each target cell. Annual NDVI smooths seasonal change and may also respond to crops and disturbance.

3.2 Grid alignment and predictor screening

Appendix A documents all 21 candidate layers with units, source period and selection status, and Appendix B maps them. Each Appendix B panel uses its own 2nd–98th percentile colour stretch. Colours therefore compare places within one layer and cannot compare values between layers.

Seven predictors were kept after a training-only screen with an absolute-correlation limit of 0.8 and a seven-variable cap. The retained layers are NDVI, elevation and five climate variables. These are BIO6 (minimum temperature of the coldest month), BIO12 (annual precipitation), BIO15 (precipitation seasonality), BIO2 (mean diurnal temperature range) and BIO14 (precipitation of the driest month). I dropped correlated layers because collinear predictors share information, which makes importance scores unstable and response curves hard to read (Dormann et al., 2013). BIO11, for example, correlated at 0.97 with a retained layer. A dropped variable leaves the fit but keeps its ecological role.

4 Model construction and evaluation

4.1 Calibration area and background

The calibration area was defined as the union of 100 km buffers around retained occurrences, clipped to India and to cells with valid predictors. I then drew 10,000 background cells uniformly from this area, excluding occurrence cells. The buffer defines the assumed

calibration area for background sampling; it is not a measured dispersal distance or an empirically verified area where the species could have been recorded. Background cells are random locations with unknown status, and uniform sampling leaves uneven survey effort in place (Barbet-Massin et al., 2012).

4.2 Model selection and test region

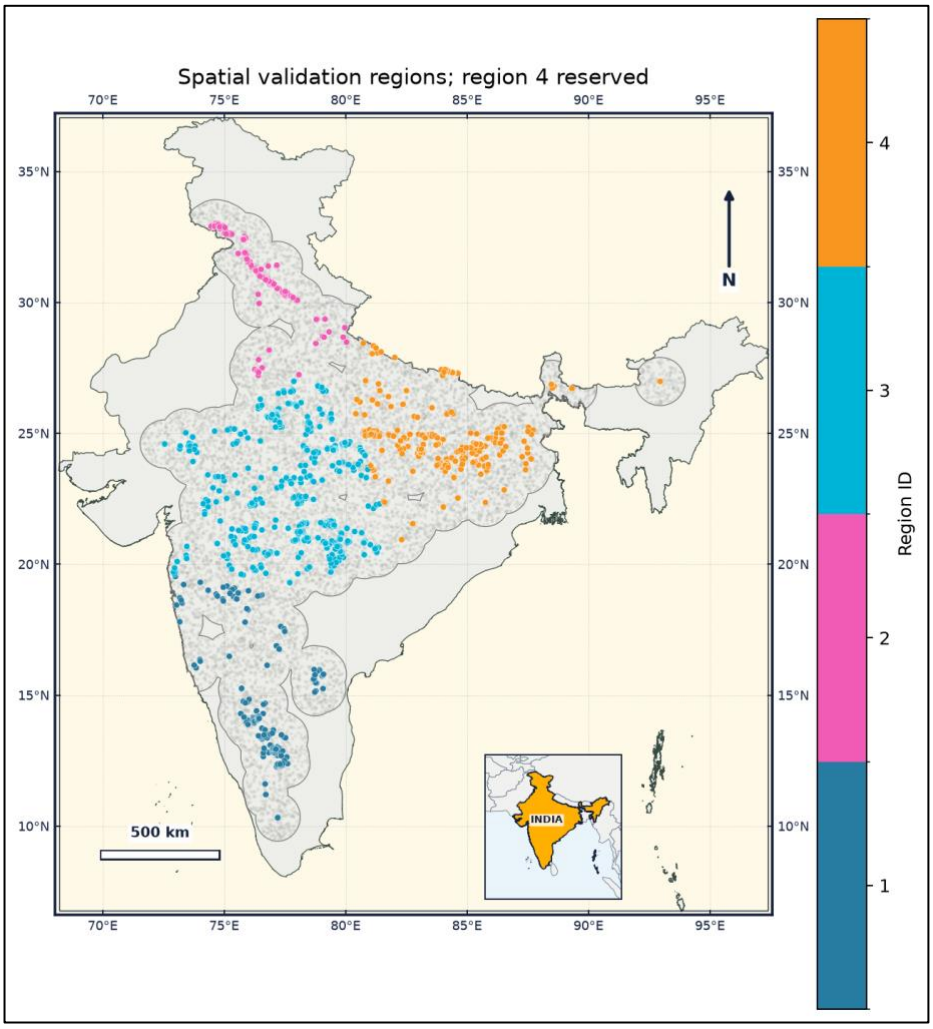


Figure 2. Geographic validation regions. Region 4 was reserved for the eastern-region test. Source: spatial assignments from the saved project run; boundary from Survey of India.

The model was fitted with the elapid package's MaxentModel, which uses an L1-regularised scikit-learn solver. I chose the settings with nested spatial cross-validation, where each fold tests on a region the model has not seen. The selected model used linear features and a regularisation multiplier of 4, which shrinks coefficients to limit overfitting. The output is a relative suitability score from 0 to 1 that ranks cells (Elith et al., 2011; Merow et al., 2013; Phillips et al., 2006, 2017; Phillips, 2021).

Occurrence cells were grouped into four geographic regions using projected coordinates and I reserved the easternmost, Region 4, to act as the validation set for the model. A 20 km exclusion halo separated test presences from training presences, and the nearest pair was 20.19 km apart. The split gave 941 development presences, 331 test presences and seven presences dropped by the halo. This design reduces spatial autocorrelation, where nearby cells look alike and inflate scores in random splits (Roberts et al., 2017). The test AUC used 331 presences and 2,800 background cells, and training used 941 presences and 7,184 background cells.

5 Results and ecological interpretation

The model ranked test-region presences above sampled background points with an AUC of 0.729 (95% interval 0.699–0.758). AUC is the probability that a random presence scores higher than a random background cell; 0.5 corresponds to random ranking for these samples. NDVI, annual precipitation and precipitation seasonality were the three most important variables. The relative-suitability surface is highest across central India, Gujarat and the Himalayan foothills, while the threshold-screened area is largest in Madhya Pradesh, Uttar Pradesh and Maharashtra.

5.1 Variables that contribute most

NDVI was the most important variable. Shuffling it cut test-region AUC by 0.154 on average over ten repeats, followed by annual precipitation (BIO12, 0.073) and precipitation seasonality (BIO15, 0.067). Elevation (0.021), BIO2 (0.012), BIO14 (0.003) and BIO6 (0.001) added little to this model's ranking. Higher scores at greener pixels are consistent with a vegetation-related signal, but NDVI is not a direct measure of woodland or species presence and can also reflect crops and disturbance. Correlated predictors share information, so shuffling one may leave a proxy in the model.

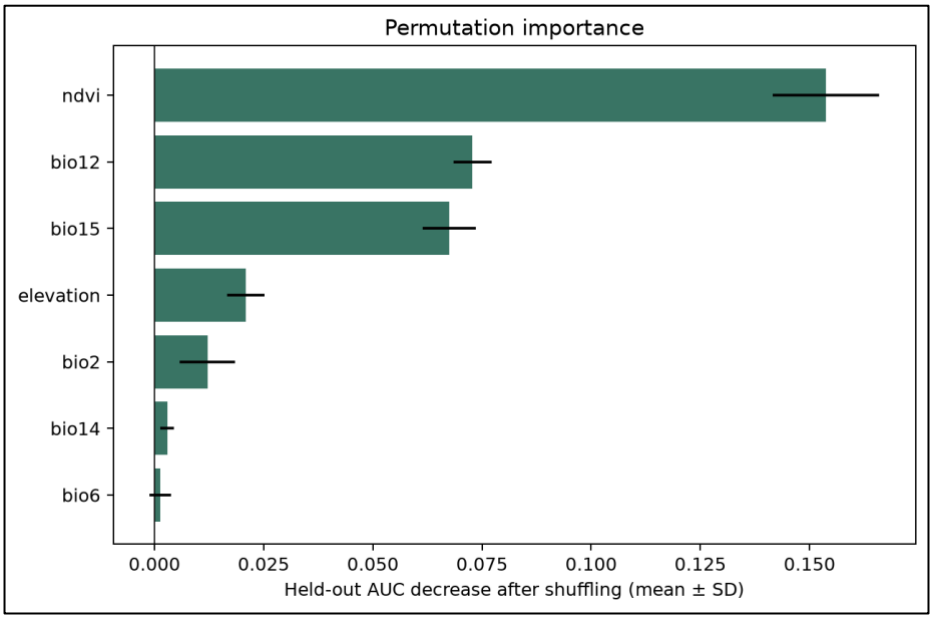


Figure 3. Permutation importance in the development-trained model, measured as test-region AUC loss over ten repeats. Source: project evaluation metrics.

The jackknife gave the same top three. Dropping NDVI lowered test-region AUC by 0.078, dropping BIO12 by 0.067 and dropping BIO15 by 0.058. NDVI alone scored 0.653 and BIO15 alone scored 0.582, both below the full model's 0.729, so no single variable matches the full model. Dropping BIO6 slightly raised AUC, which suggests BIO6 adds noise in the test region. Both tests measure how the fitted score depends on each input and cannot show cause.

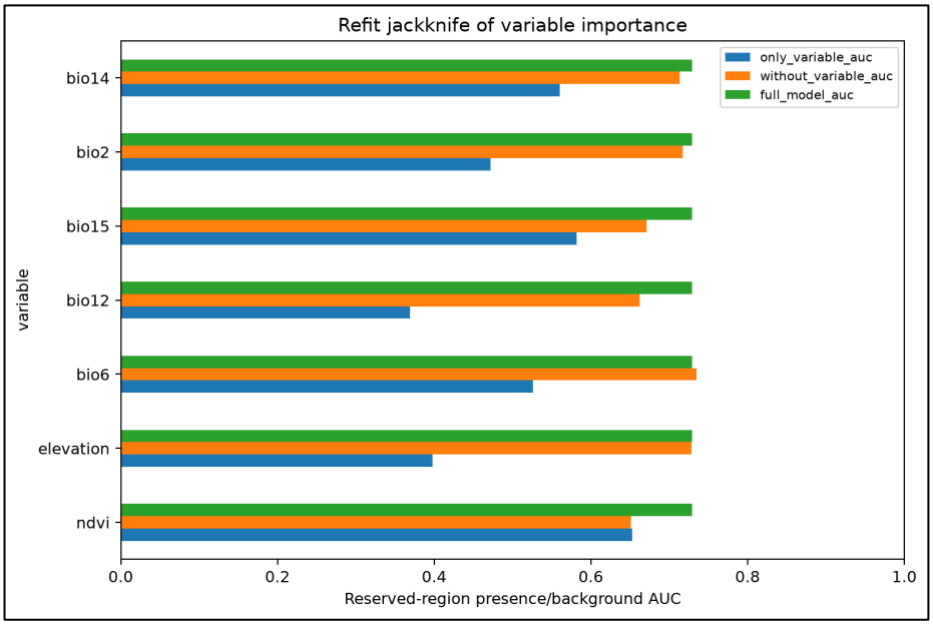


Figure 4. Test-region jackknife refits. Bars show AUC with one predictor alone, with one predictor removed, and for the full model. Source: project evaluation metrics.

Figure 5 shows response curves from the final full-data model. The importance scores come from the earlier development model, so their sizes can differ from the curves.

Suitability rises steeply with NDVI, by 0.87 from the lowest to the highest values on the curve. NDVI represents vegetation cover and productivity, so the species scores highest in green, wooded land. Annual NDVI also responds to crops, so this link is only a hint about natural habitat.

Suitability falls with annual precipitation (change of −0.55) and rises with precipitation seasonality (+0.60) and driest-month precipitation (+0.39). BIO12 reflects total water supply, BIO15 reflects how sharply rain concentrates in one season, and BIO14 reflects dry-season water. Together they fit a deciduous tree of seasonally dry climates. BIO12 and BIO14 pull in opposite directions because each curve is conditional on the other predictors, and sampling may also play a role.

Suitability falls with elevation (−0.35) and diurnal temperature range, BIO2 (−0.42), while the BIO6 curve is nearly flat. The low BIO6 importance and small response change indicate little unique ranking information from coldest-month temperature in this fitted model; they do not establish that temperature is ecologically irrelevant. Elevation and BIO2 may also track moisture, land-use and access gradients.

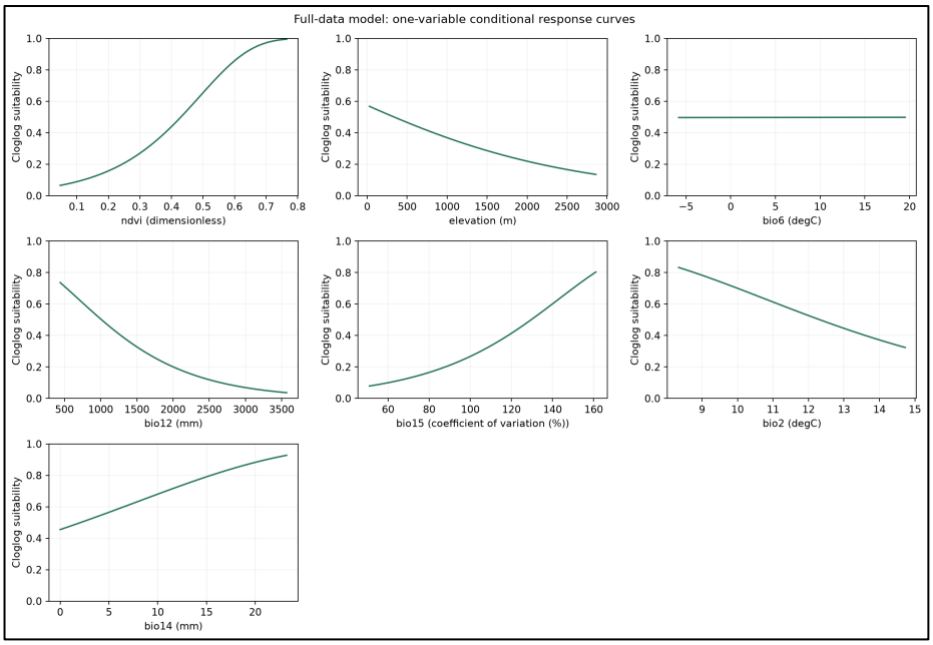


Figure 5. One-variable response curves from the final full-data model. Source: saved project run.

5.2 AUC, strengths, weaknesses and overfitting

AUC summarises ranking across all thresholds, so it compares models without choosing one cutoff. It has three weaknesses here. It depends on the background I sampled. It does not test calibration. It cannot give a false-positive rate, because background cells are not known absences (Fielding & Bell, 1997; Lobo et al., 2008; Phillips et al., 2006).

The model's test-region AUC was 0.729, with a 95% interval of 0.699–0.758 from 2,000 bootstrap resamples. A distance-only baseline scored 0.554, so the environmental model's AUC was 0.175 higher in this evaluation. No significance test was performed on that difference. The result indicates moderate ranking discrimination for this reserved region and background design; it does not establish calibrated occupancy probabilities or guarantee transfer to other regions. The interval reflects resampling variation, while spatial sorting and other sources of uncertainty remain.

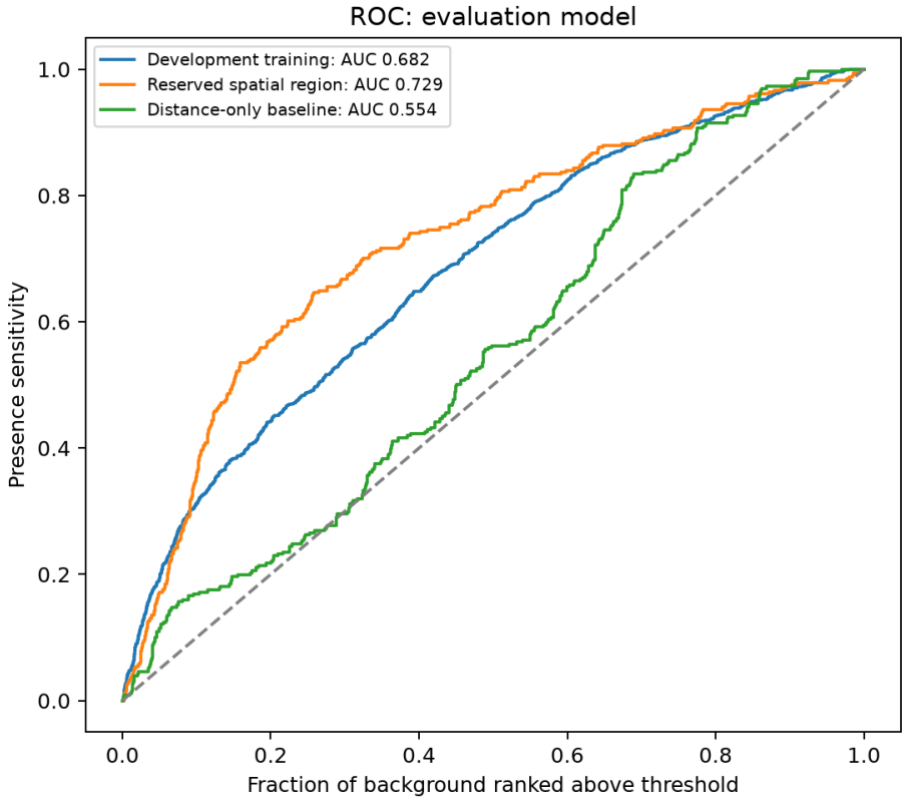


Figure 6. Receiver-operating-characteristic curves for development training, the test region and a distance-only baseline. AUC compares presence scores with sampled background scores.

Training AUC was 0.682 and test-region AUC was 0.729, so this comparison does not show the usual pattern of training performance exceeding test performance. It does not rule out overfitting. The learning curve rose from 0.714 to 0.729 in held-out AUC as training presences

grew from 235 to 941, while the three spatial development folds varied from 0.590 to 0.754 (mean 0.690, SD 0.087). The weakest fold and variation across regions show that transfer performance is uneven.

When the model was trained on other-species survey locations as background, the same test region scored 0.431. This shows the score depends strongly on how the background represents observation bias, a sampling-bias problem from the course. This alternative run was not retuned, so read it as a warning about sensitivity and leave a full comparison to future work.

At the development T10 threshold of 0.344, the model missed 3.0% of test presences (interval 1.2–5.1%). T10 is the score that 90% of training presences exceed. Yet 89.5% of test background cells also scored above it, so the threshold is broad and keeps most background along with nearly all presences. Low omission alone does not justify a binary habitat map, and the 89.5% is not a false-positive rate.

5.3 What the map shows

The map shows a patchy surface of relative suitability that follows vegetation, rainfall and terrain (Figure 7). Higher scores appear across central India, Gujarat, the Himalayan foothills and parts of the northeast; lower scores appear in the high Himalaya, western desert and interior Deccan plateau. These are modelled associations within the valid-data footprint, not confirmed habitat boundaries. The pattern is broadly consistent with descriptions of the species' seasonal environments, but survey bias and planted records remain possible explanations for part of the pattern. A high score can help select a cell for survey; field observations are needed to establish presence.

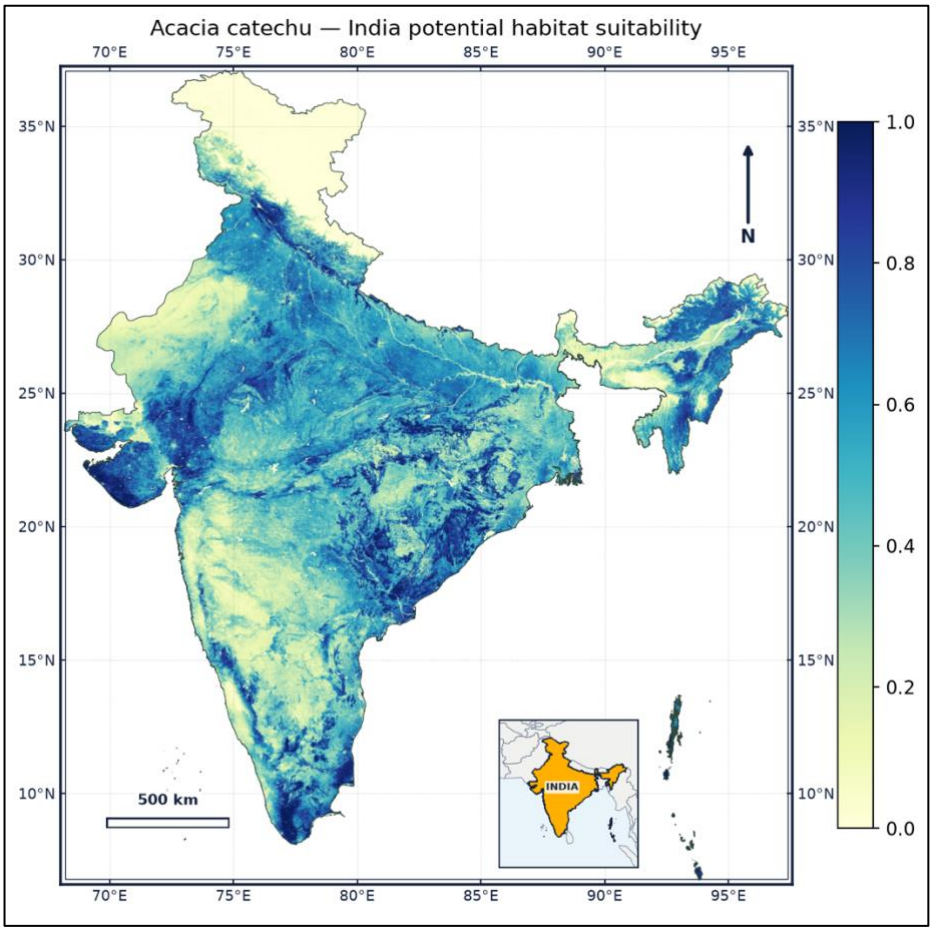


Figure 7. Relative suitability from the final full-data model. Colours show the relative suitability score. Source: saved model and clipped environmental layers; boundary from Survey of India.

Of 4,176,823 valid raster cells, 2,056,104 lie inside the calibration area and within the training ranges. Another 1,663,105 lie outside the calibration area but within the ranges, and 457,614 are environmentally novel (Figure 8). Predictions have the most support in the first group and the least in novel cells. Clamping restricts predictor values to the training range; it does not validate predictions in novel environments.

The final model has an apparent training AUC of 0.695 and a T10 threshold of 0.328. Applying that threshold inside the calibration area and within the training predictor ranges identifies approximately 1,454,463 km² of threshold-screened area. The largest state totals are Madhya Pradesh (265,539 km²), Uttar Pradesh (225,446 km²), Maharashtra (158,223 km²), Rajasthan (143,641 km²) and Chhattisgarh (80,464 km²). These totals describe model-screened grid-cell area under the stated calibration and threshold rules; they are not estimates of occupied land or confirmed habitat. The threshold is broad, so the area should be interpreted as a screening result, not a mapped range.

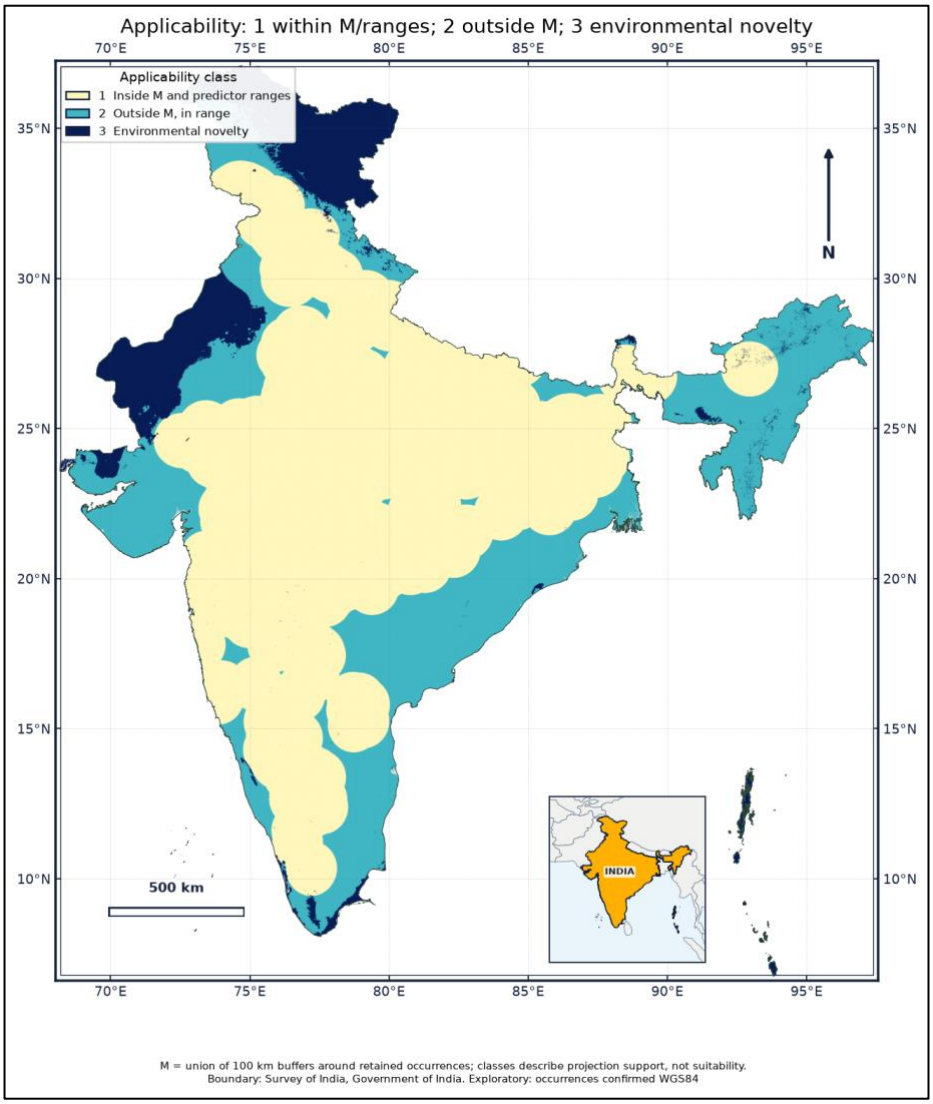


Figure 8. Projection applicability classes: inside the calibration area and predictor ranges, outside the calibration area but within ranges, and environmentally novel. Source: saved project run.

6 Implications for conservation and risk

6.1 Conservation planning

The model is most useful for planning surveys. Madhya Pradesh has the largest area above the threshold (265,539 km²). A team there could survey high-, mid- and low-score cells across land-cover types and levels of road access. Detections in low-score cells would show where the ranking fails, and non-detections in high-score cells would show where it overpredicts. Recording effort and non-detections as carefully as detections makes both outcomes usable. A high score cannot confirm a viable population, and protected-area benefit needs a separate overlay and field data.

6.2 Climate-change studies

This model describes current conditions and has no future projection. Annual precipitation and precipitation seasonality rank among the top variables, so a shift in rainfall would move the map most. For a climate study I would swap in WorldClim future layers for 2041–2060 under SSP2-4.5 and SSP5-8.5, and rerun the novelty check. NDVI has no future version, so I would compare a climate-only refit with the current model. The mismatch between 1970–2000 climate and 2025 NDVI needs fixing first.

6.3 Invasive-species risk

A suitability surface shows where conditions resemble known records, and invasion risk needs more evidence. Consider a forest department planning to plant *A. catechu* on degraded land in Rajasthan, where 143,641 km² lies above the threshold. The map gives a first climate check for that plan. The department would still need records that separate native, planted, naturalised and invasive stands, plus data on dispersal and disturbance, before judging spread. The India-wide surface gives no evidence that the species is invasive anywhere in India.

7 Limitations and next steps

The main limit is the observation process. The workbook gives no consistent dates, survey effort, coordinate uncertainty or wild-versus-planted status. Cell thinning removes duplicate records but leaves road access and collection intensity in the data. The uniform background and the 100 km buffer are my choices, and the 0.431 score under another background shows they need testing.

The model pairs 1970–2000 climate with 2025 NDVI, omits land use and disturbance, and tests transfer on one test region and three development folds. The NDVI layer is evaluated at 1 km and skips explicit averaging of 10 m pixels. I ran no significance test on the AUC gap over the distance-only baseline. The threshold and area figures depend on the T10 rule, grid size, calibration area and predictors.

Next, I would verify taxonomy and wild or planted status, record dates and coordinate precision, and justify the accessible area from the observation process. I would retune alternative backgrounds inside the same nested spatial design, add a time-matched vegetation layer, and validate with spatially balanced field data. Any public map should carry its applicability mask and state that it shows relative suitability and gives no confirmed occupancy.

8 Conclusion

Spatially blocked evaluation indicates that the model contains environmental ranking signal for Acacia catechu in the reserved eastern region (AUC = 0.729; 95% interval = 0.699–0.758). That result is qualified by variation among spatial folds (mean AUC = 0.690, SD = 0.087), a weak fold at 0.590, and a sensitivity score of 0.431 under the other-species background. The prediction surface is therefore best used to guide field surveys, not as an occupancy estimate or confirmed national distribution. NDVI, annual precipitation and precipitation seasonality carried the strongest model-dependent signal, but these associations do not establish ecological causes. The threshold-screened area of approximately 1,454,463 km² is conditional on the calibration region, predictor ranges and T10 rule; it is not occupied area. The next priority is independent, effort-recorded field validation across contrasting score classes, supported by better records of dates, positional uncertainty and wild or planted status. Conservation, planting and climate-planning decisions should not rely on this map alone.

References

Barbet-Massin, M., Jiguet, F., Albert, C. H., & Thuiller, W. (2012). Selecting pseudo-absences for species distribution models: How, where and how many? *Methods in Ecology and Evolution*, *3*(2), 327–338. https://doi.org/10.1111/j.2041-210X.2011.00172.x

Deb, D., Singh, J. P., & Chaurasia, R. S. (2015). Biomass estimation of Acacia catechu using statistical modelling. *Range Management and Agroforestry*, *36*(2), 151–155. https://publications.rmsi.in/index.php/rma/article/view/341

Dormann, C. F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., García Márquez, J. R., Gruber, B., Lafourcade, B., Leitão, P. J., Münkemüller, T., McClean, C., Osborne, P. E., Reineking, B., Schröder, B., Skidmore, A. K., Zurell, D., & Lautenbach, S. (2013). Collinearity: A review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, *36*(1), 27–46. https://doi.org/10.1111/j.1600-0587.2012.07348.x

Elith, J., & Leathwick, J. R. (2009). Species distribution models: Ecological explanation and prediction across space and time. *Annual Review of Ecology, Evolution, and Systematics*, *40*, 677–697. https://doi.org/10.1146/annurev.ecolsys.110308.120159

Elith, J., Phillips, S. J., Hastie, T., Dudík, M., Chee, Y. E., & Yates, C. J. (2011). A statistical explanation of MaxEnt for ecologists. *Diversity and Distributions*, *17*(1), 43–57. https://doi.org/10.1111/j.1472-4642.2010.00725.x

Fielding, A. H., & Bell, J. F. (1997). A review of methods for the assessment of prediction errors in conservation presence/absence models. *Environmental Conservation*, *24*(1), 38–49. https://doi.org/10.1017/S0376892997000088

Fick, S. E., & Hijmans, R. J. (2017). WorldClim 2: New 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, *37*(12), 4302–4315. https://doi.org/10.1002/joc.5086

Google. (n.d.). Harmonized Sentinel-2 MSI: Multispectral instrument, Level-2A (SR) [Data set]. *Google Earth Engine Data Catalog* https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED

Google Earth Engine. (2026). Sentinel-2 Cloud Score+ harmonized archive and quality bands [Data set]. *Google Earth Engine Data Catalog* https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_CLOUD_SCORE_PLUS_V1_S2_HARMONIZED

Grubb, P. J. (1977). The maintenance of species-richness in plant communities: The importance of the regeneration niche. *Biological Reviews*, *52*(1), 107–145. https://doi.org/10.1111/j.1469-185X.1977.tb01347.x

Hijmans, R. J. (2012). Cross-validation of species distribution models: Removing spatial sorting bias and calibration with a null model. *Ecology*, *93*(4), 679–688. https://doi.org/10.1890/11-0826.1

Indian Council of Forestry Research and Education. (n.d.). Senegalia catechu (L.) P.J.H. Hurter & Mabb. Forestry species information. https://www.icfre.gov.in/pdf/books-file/book27.pdf

Jha, M. N. (1995). Seed germination studies in Acacia catechu Willd. *Indian Forester*, *121*(6), 513–517.

Jhilta, P., Rana, N., Dipta, B., & Ghabru, A. (2018). Studies on the pod, seed and germination traits of Acacia catechu from Uttarakhand. *International Journal of Zoology Studies*, *3*(4), 42–46. https://www.zoologyjournals.com/assets/archives/2018/vol3issue4/3-4-21-681.pdf

Kew Science. (2026). Senegalia catechu (L.f.) P.J.H. Hurter & Mabb. *Plants of the World Online* https://powo.science.kew.org/taxon/urn:lsid:ipni.org:names:77089268-1

Lobo, J. M., Jiménez-Valverde, A., & Real, R. (2008). AUC: A misleading measure of the performance of predictive distribution models. *Global Ecology and Biogeography*, *17*(2), 145–151. https://doi.org/10.1111/j.1466-8238.2007.00358.x

Merow, C., Smith, M. J., & Silander, J. A. (2013). A practical guide to MaxEnt for modeling species' distributions: What it does, and why inputs and settings matter. *Ecography*, *36*(10), 1058–1069. https://doi.org/10.1111/j.1600-0587.2013.07872.x

Negi, S. S., Tewari, J. C., & Singh, S. (1996). Natural regeneration of Acacia catechu in dry tropical forests of northern India. *Journal of Tropical Forest Science*, *8*(4), 494–503.

Orwa, C., Mutua, A., Kindt, R., Jamnadass, R., & Anthony, S. (2009). Acacia catechu. Agroforestree Database: A tree reference and selection guide (Version 4.0). https://apps.worldagroforestry.org/treedb/AFTPDFS/Acacia_catechu.PDF

Phillips, S. J. (2021). A brief tutorial on Maxent. https://biodiversityinformatics.amnh.org/open_source/maxent/

Phillips, S. J., Anderson, R. P., & Schapire, R. E. (2006). Maximum entropy modeling of species geographic distributions. *Ecological Modelling*, *190*(3–4), 231–259. https://doi.org/10.1016/j.ecolmodel.2005.03.026

Phillips, S. J., Anderson, R. P., Dudík, M., Schapire, R. E., & Blair, M. E. (2017). Opening the black box: An open-source release of Maxent. *Ecography*, *40*(7), 887–893. https://doi.org/10.1111/ecog.03049

Puri, S., Swamy, S. L., & Jaiswal, A. K. (1994). Water relations and photosynthetic characteristics of seedlings of four dry zone tree species. *Indian Journal of Forestry*, *17*(3), 185–192.

Raizada, A., & Juyal, G. P. (2012). Tree species diversity, species regeneration and biological productivity of seeded Acacia catechu Willd. in rehabilitated limestone mines in the north-west Indian Himalayas. *Land Degradation & Development*, *23*(2), 167–174. https://doi.org/10.1002/ldr.1067

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, *40*(8), 913–929. https://doi.org/10.1111/ecog.02881

Singh, G., Joshi, R. D., Chopra, S. K., & Singh, A. B. (1974). Late Quaternary history of vegetation and climate of the Rajasthan desert, India. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *267*(889), 467–501. https://doi.org/10.1098/rstb.1974.0006

Sourabh, P., Patel, R., & Singh, V. (2023). Predicting potential habitat distribution of Acacia catechu in India. *Applied Ecology and Environmental Sciences*, *11*(3), 99–103. https://doi.org/10.12691/aees-11-3-4

Survey of India. (n.d.). Administrative Boundary Data Base (ABDB). https://surveyofindia.gov.in/pages/administrative-boundary-data-base-abdb-

WorldClim. (2026). WorldClim version 2.1 climate data. https://www.worldclim.org/data/worldclim21.html

Appendix A Predictor table

Table A1 gives the source and period of each layer. “Selected” marks the seven predictors in the final model. The training-only priority, correlation and predictor-cap rules removed the other layers.

Table A1. Candidate environmental layers clipped and aligned for this analysis. Climate layers: WorldClim 2.1, 1970–2000. Elevation: SRTM-derived. NDVI: Sentinel-2 annual mean, 2025. All layers: 30 arc-seconds, EPSG:4326. Source: project environmental-variable register; WorldClim and Google Earth Engine dataset documentation.

Layer	Variable definition	Units	Source period	Selected
BIO1	Annual mean temperature	°C	1970–2000	No
BIO2	Mean diurnal temperature range	°C	1970–2000	Yes
BIO3	Isothermality	%	1970–2000	No
BIO4	Temperature seasonality	temperature SD × 100	1970–2000	No
BIO5	Maximum temperature of warmest month	°C	1970–2000	No
BIO6	Minimum temperature of coldest month	°C	1970–2000	Yes
BIO7	Temperature annual range	°C	1970–2000	No
BIO8	Mean temperature of wettest quarter	°C	1970–2000	No
BIO9	Mean temperature of driest quarter	°C	1970–2000	No
BIO10	Mean temperature of warmest quarter	°C	1970–2000	No
BIO11	Mean temperature of coldest quarter	°C	1970–2000	No
BIO12	Annual precipitation	mm	1970–2000	Yes
BIO13	Precipitation of wettest month	mm	1970–2000	No
BIO14	Precipitation of driest month	mm	1970–2000	Yes
BIO15	Precipitation seasonality	coefficient of variation (%)	1970–2000	Yes
BIO16	Precipitation of wettest quarter	mm	1970–2000	No
BIO17	Precipitation of driest quarter	mm	1970–2000	No
BIO18	Precipitation of warmest quarter	mm	1970–2000	No
BIO19	Precipitation of coldest quarter	mm	1970–2000	No
ELEVATION	SRTM-derived elevation	m	static terrain	Yes
NDVI	Mean Sentinel-2 NDVI	dimensionless	January–December 2025 annual mean	Yes

Appendix B Clipped predictor maps

The atlas covers the 19 bioclimatic layers, SRTM-derived elevation and 2025 annual mean NDVI, clipped to India. Each panel uses its own 2nd–98th percentile stretch, so colours are relative within a panel and carry no shared numeric legend across variables. I aligned all rasters to the same 30-arc-second EPSG:4326 grid and clipped them to the study boundary.

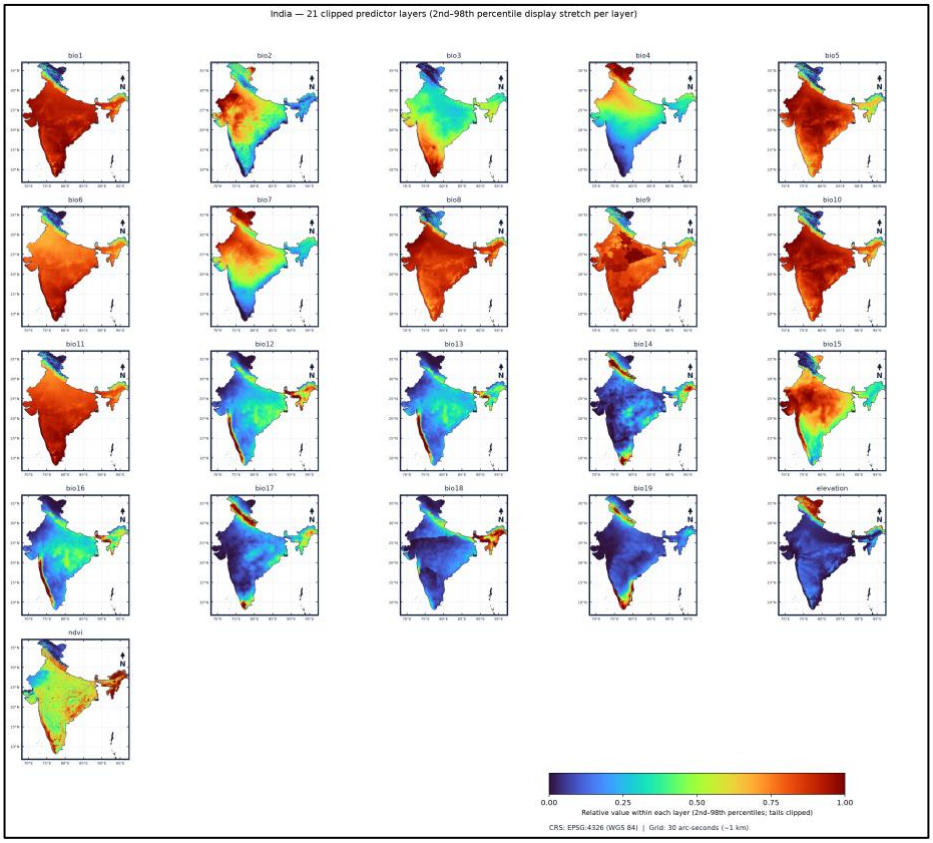


Figure B1. Atlas of all 21 clipped environmental predictor layers. Individual display stretches aid visual inspection; units are listed in Table A1. Source: WorldClim 2.1, SRTM-derived elevation, Sentinel-2 processed in Google Earth Engine, and Survey of India boundary.

Appendix C Supplementary generated figures

The following figures are supplementary diagnostics and input documentation generated by the saved Python workflow. They complement, rather than replace, the maps and diagnostics discussed in the main text.

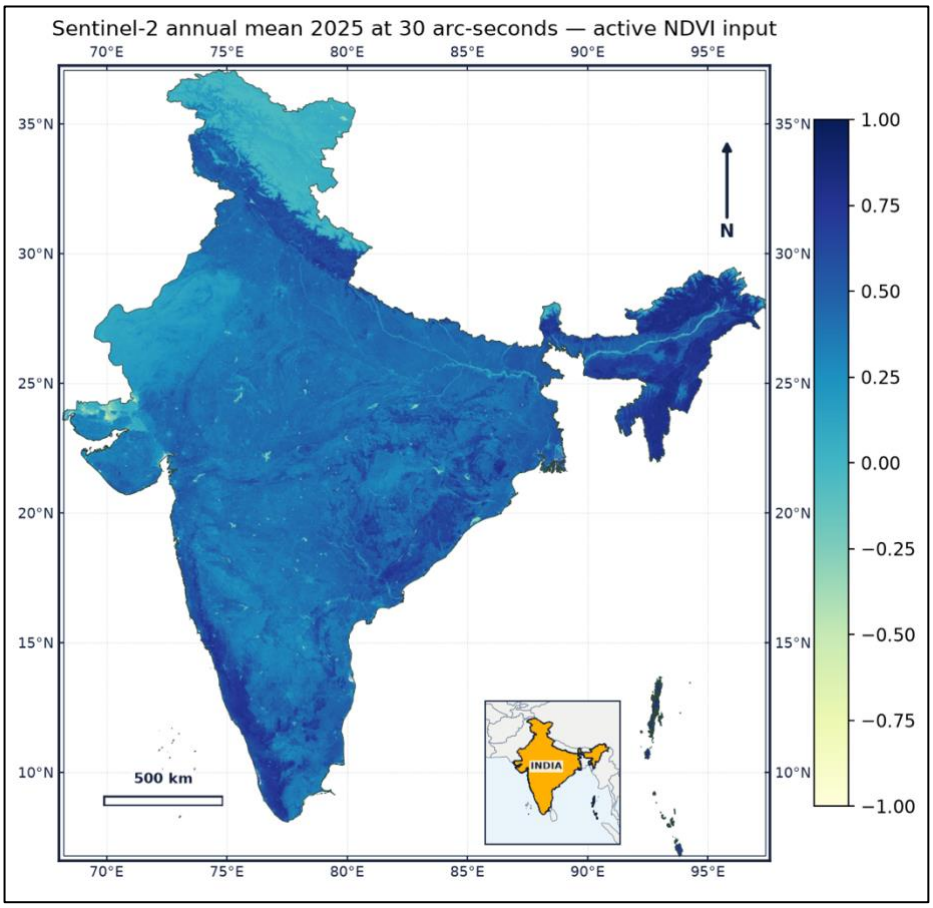


Figure C1. Sentinel-2 annual mean NDVI input for 2025 on the analysis grid. Source: saved model and clipped environmental layers; boundary from Survey of India.

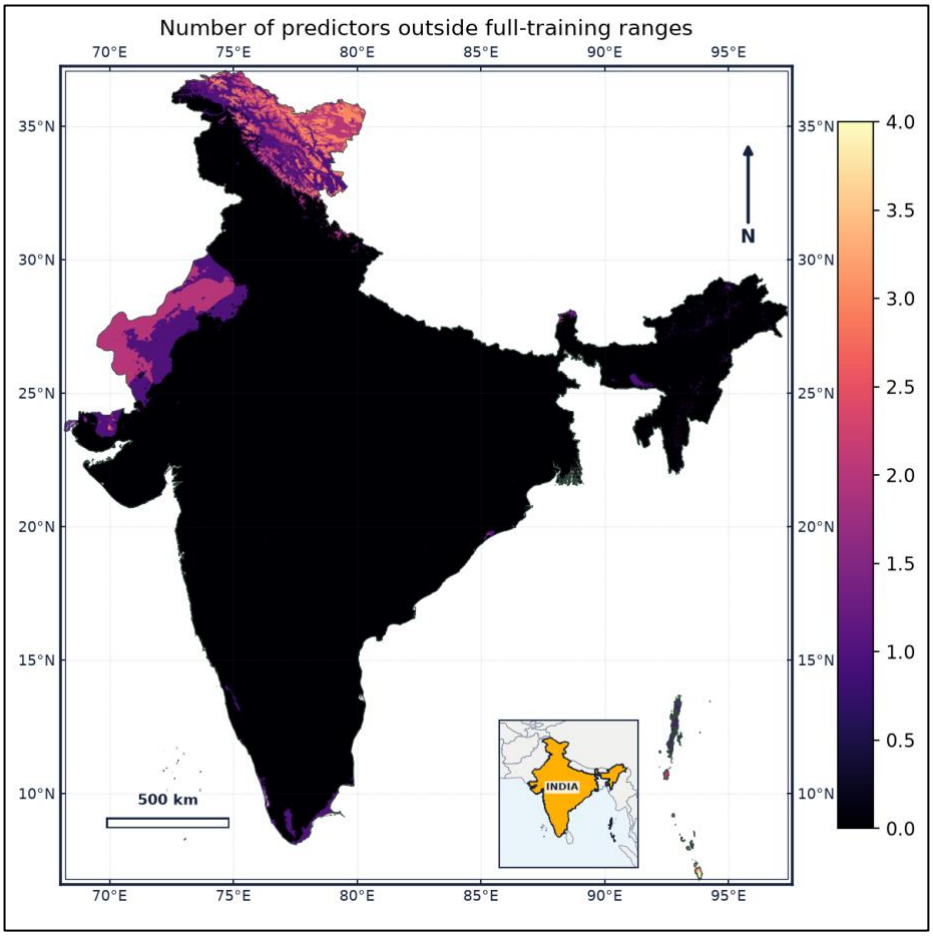


Figure C2. Count of predictors outside their full-training ranges. Novelty is an applicability diagnostic, not a suitability score. Source: saved model and clipped environmental layers; boundary from Survey of India.

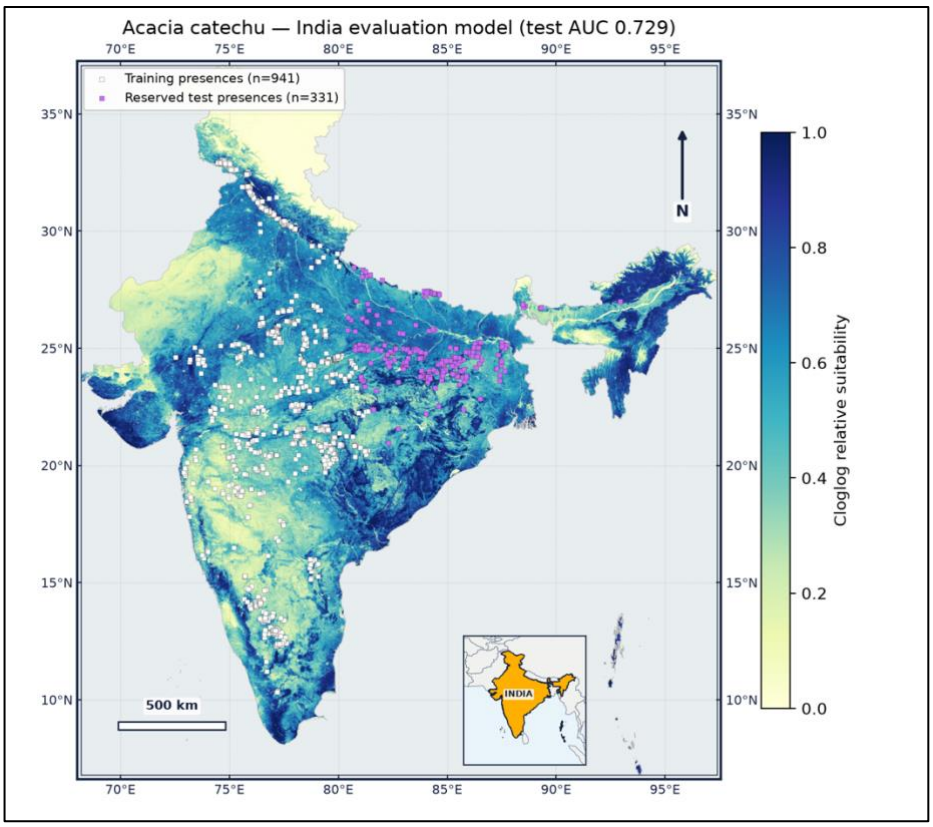


Figure C3. Evaluation-model suitability surface with development and held-out occurrence points. Scores are relative, not occupancy probabilities.

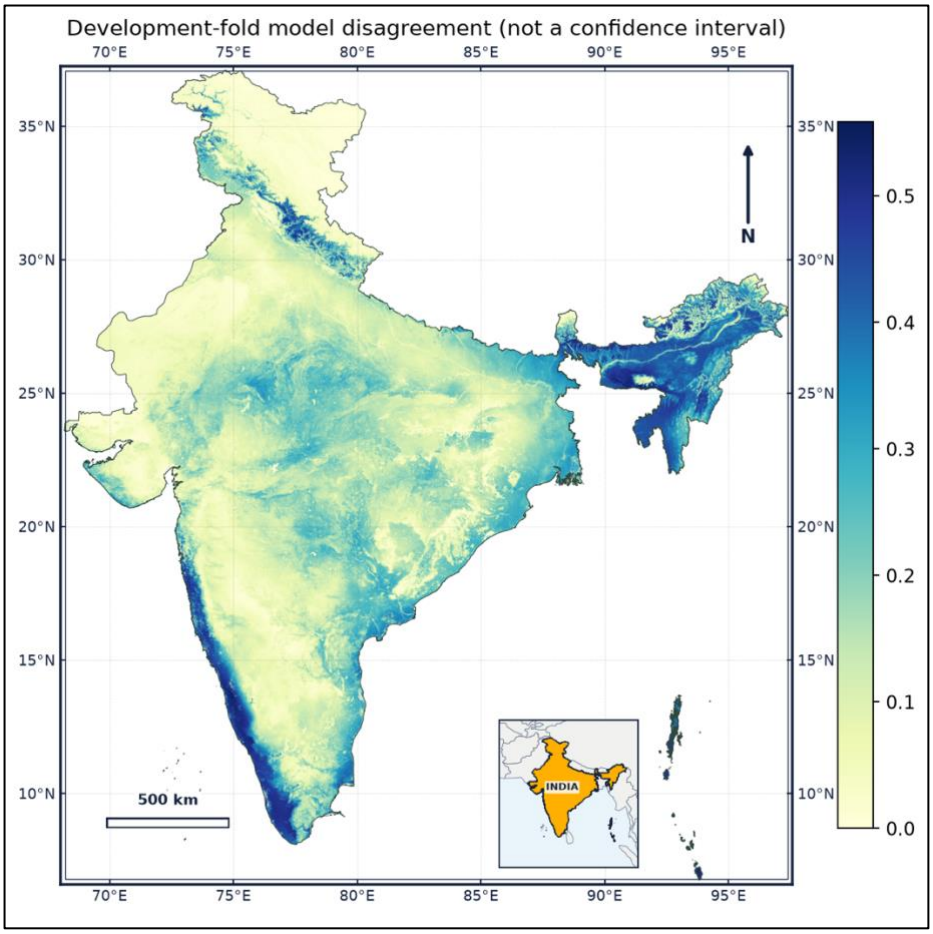


Figure C4. Standard deviation of predictions across development-fold models. This is fold disagreement, not a confidence interval.

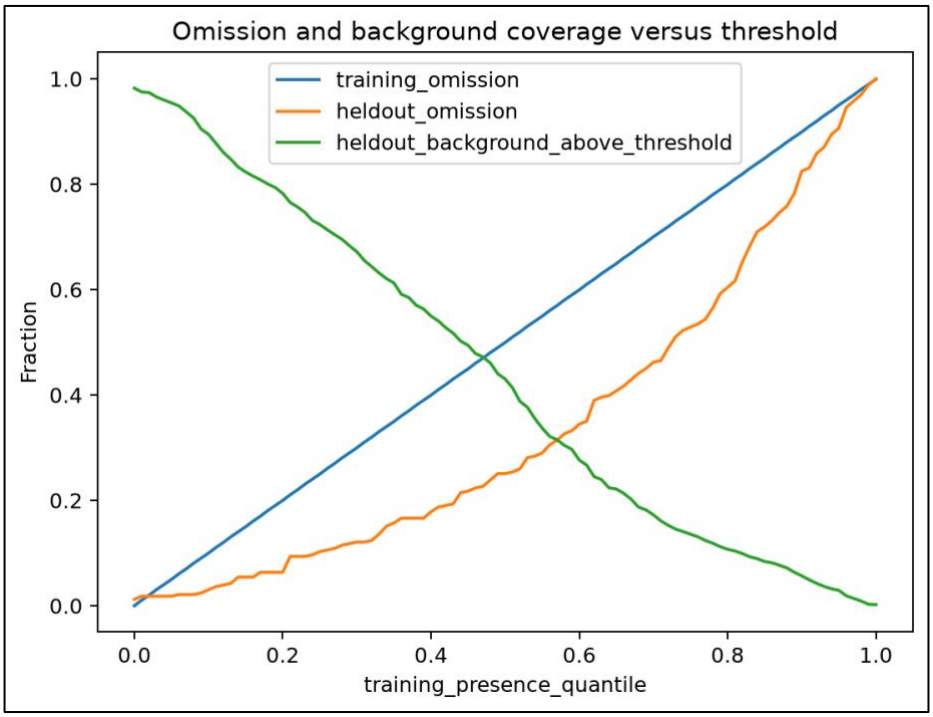


Figure C5. Training omission, held-out omission and held-out background coverage across thresholds.

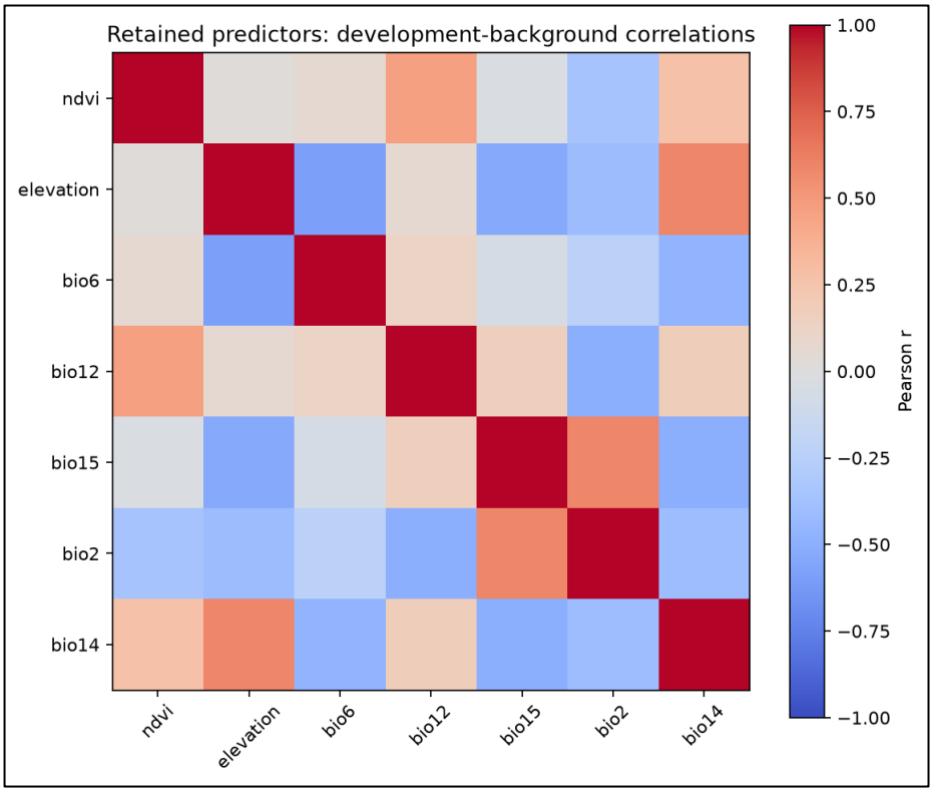


Figure C6. Pairwise correlations among the seven retained predictors in the development-background data.